

Continuation of Weekly Plan for Course Content	
Weeks	Topics
7	Allocated Experiments (1-6) to be performed by Section A2/B2 in Phy Lab
8	Allocated Experiments (1-6) to be performed by Section A1/B1 in Phy Lab
9	Allocated Experiments (1-6) to be performed by Section A2/B2 in Phy Lab
10	Allocated Experiments (1-6) to be performed by Section A1/B1 in Phy Lab
11	Allocated Experiments (1-6) to be performed by Section A2/B2 in Phy Lab
12	Allocated Experiments (1-6) to be performed by Section A1/B1 in Phy Lab
13	Allocated Experiments (1-6) to be performed by Section A2/B2 in Phy Lab
14	Sessional Quiz/ Lab Test
15	Allocated Experiments (1-6) to be performed by Section A/B in Phy Lab (Make up)

Assessment Strategy

- Lab Tasks
- Lab Test / Quiz

Distribution of Marks

- Attendance: 10%
- Lab Tasks: 70%
- Lab Test/Quiz: 20%

Recommended Texts

- Physics Practical by Prof. Giasuddin and Shahabuddin

1.5.1.8 Description of CSE 4203

General Information

- **Course Title:** Discrete Mathematics
- **Type of Course:** Theory and Compulsory

Course Details

Course Contents (As approved by the Academic Council)

Set theory, Elementary number theory, Graph theory, Paths and trees, Generating functions, Algebraic structures, Semigraph, Permutation groups, Binary relations, functions, Mathematical logic, Propositional calculus and predicate calculus.

Course Objectives

- Develop logical thinking and problem-solving skills applicable to computer science and engineering.
- Apply mathematical and theoretical computer science concepts to structure, analyze, and solve complex computational problems.

- Enhance the ability to reason abstractly and critically, understanding the applicability and limitations of formal methods.

Knowledge Required

None

Course Outcomes

COs	CO Statements	Corresponding POs	Learning Domain and Taxonomy Levels	Delivery Methods and Activities	Assessment Tools
CO1	Interpret and apply propositional logic, predicate logic, rule of inference, and proof strategy.	PO1	C2, C3	Lecture	Quiz, Mid Semester Examination, Semester Final Examination
CO2	Interpret and apply the discrete structures such as sets, functions, sequence and summation. Algorithm, growth of functions, complexity analysis of algorithms, and relation.	PO1	C2, C3		
CO3	Interpret and apply number theory and counting principles for combinatorial analysis.	PO1	C2, C3		
CO4	Interpret and apply graphs, and trees for modeling discrete objects; demonstrate this knowledge and solve realistic problems.	PO1	C2, C3		

* For details of program outcome (PO) statements, please check section A.2, and for details of domains and taxonomy level, please check section A.3.

Mapping of Knowledge Profile, Complex Engineering Problem Solving and Complex Engineering Activities

K1	K2	K3	K4	K5	K6	K7	K8	P1	P2	P3	P4	P5	P6	P7	A1	A2	A3	A4	A5
	✓	✓	✓					✓											

Lecture Plan

Weekly Plan for Course Content	
Weeks	Topics
1	Introduction to propositional logic.
2	Application of Propositional Logic Propositional Equivalence, Predicates and Quantifiers
3	Rules of Inference, introduction to Proof Techniques.
4	Sets, Set Operations.

Continuation of Weekly Plan for Course Content	
Weeks	Topics
5	Functions, Sums and Sequences and Relation.
6	Algorithms, Searching and Sorting Algorithms, Halting Problem.
7	Growth of Functions, Complexity of Algorithms
8	Midterm examinations
9	Number Theory Basics, Divisibility and Modular Arithmetic, Base Conversion , Algorithms for Integer Operations (Addition, Multiplication, Division)
10	Number Theory Applications, Primes Numbers, Pseudorandom Numbers, Cryptography.
11	Induction and Recursion.
12	Introduction to Graph Theory, Graph definitions, types of graphs, graph representation (adjacency matrix, adjacency list), basic graph terminology.
13	Advanced Graph Concepts, Paths, cycles, connectivity, Eulerian and Hamiltonian graphs, planar graphs, graph algorithms (BFS, DFS), graph applications.
14	Trees, Tree definitions, rooted trees, binary trees, spanning trees, applications in computing and algorithms.
15	Comprehensive Review.
16	Semester Final Examination

Assessment Strategy

- Quiz
- Mid Semester Examination
- Semester Final

Distribution of Marks

- 20%
- 40%
- 40%

Recommended Texts

- Discrete mathematics and its applications, 8th Edition by Kenneth H. Rosen.
- Discrete mathematics with applications by Susanna S. Epp.

1.5.1.9 Description of CSE 4205

General Information

- **Course Title:** Digital Logic Design
- **Type of Course:** Theory and Compulsory